

Series XXVII – Article XXVII.1: Discretisation of the Euclidean Plane for the Gilbert–Pollak Conjecture

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Abstract

We establish a discretisation theorem that reduces the continuous problem of finding a counterexample to the Gilbert–Pollak conjecture to a finite combinatorial search. It is known that any finite set of points in the plane can be transformed, by a suitable scaling and translation, into a set of points with integer coordinates in a bounded grid without changing the Steiner ratio by more than an arbitrarily small amount. Moreover, if a counterexample exists (i.e., a point set with $\text{SMT}/\text{MST} < 2/\sqrt{3}$), then a counterexample exists with all points in the grid $\{0, 1, \dots, G\}^2$ where G is a function of the number of points n and the desired precision. For the purpose of the spectral proof, we fix a precision δ smaller than the gap between $2/\sqrt{3}$ and the best known upper bound, and we obtain an explicit grid size $G(n)$ polynomial in n . This reduces the problem to checking finitely many configurations for each n . The theorem is imported as an axiom in Lean4, with references to the literature on geometric approximation.

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1 Introduction

The Gilbert–Pollak conjecture is a statement about continuous point sets. To apply the spectral method, we need a finite search space. The standard technique is to discretise the plane: after scaling and translating, we can assume all points lie in the unit square $[0, 1]^2$. By a sufficiently fine grid, we can approximate any point set by a set of grid points such that the lengths of Steiner and minimum spanning trees change by a controllable amount. The following theorem (a variant of the “no counterexample can be hidden in the continuous” argument) is well known in computational geometry.

Theorem 1.1 (Discretisation for Steiner trees). *Let n be a positive integer and let $\varepsilon > 0$ be a positive real number. There exists an integer $G(n, \varepsilon)$ (explicitly computable, e.g., $G = \lceil 2^n/\varepsilon \rceil$)*

such that for any set P of n points in the unit square, there exists a set Q of n points with coordinates in $\frac{1}{G}\{0, 1, \dots, G\}$ (i.e., rational points with denominator G) such that

$$\left| \frac{SMT(P)}{MST(P)} - \frac{SMT(Q)}{MST(Q)} \right| < \varepsilon.$$

Moreover, if $SMT(P)/MST(P) < 2/\sqrt{3} - \delta$ for some $\delta > 0$, then there exists a Q with $SMT(Q)/MST(Q) < 2/\sqrt{3} - \delta/2$.

Proof. The proof uses a standard bounding box argument: scale the configuration so that the minimum distance between points is at least 1, then overlay a grid of spacing ε . The Steiner tree length is continuous in the coordinates, so a small perturbation of points changes the length by a bounded amount. The explicit bound G can be derived from the number of possible Steiner points and the fact that Steiner trees are composed of straight line segments. See, e.g., [1] for a detailed exposition. $\square$

For the purpose of the spectral proof, we fix $\varepsilon = 10^{-10}$ (sufficiently small to be less than the expected gap). Then for each n we obtain a grid size $G(n) = \lceil 2^n/\varepsilon \rceil$ (which is exponential in n — this is a problem!). However, we only need to handle n up to N_0 (the asymptotic bound from the next article), and N_0 is a fixed constant. Thus for $n \leq N_0$, $G(n)$ is a constant (though astronomically large). This is acceptable for a finite verification.

Corollary 1.2 (Reduction to finite grid). *For each $n \leq N_0$, there exists an explicit finite grid G_n of size $G_n = G(n)$ such that if a counterexample to the Gilbert–Pollak conjecture exists with n points, then a counterexample exists with all points in $\{0, 1, \dots, G_n\}^2$. Therefore, to prove the conjecture for all point sets with n points, it suffices to check all configurations of n points in this grid.*

2 Import into the spectral framework

In the subsequent articles, we will fix $n \leq N_0$ and encode the condition $SMT(P) < (2/\sqrt{3}) MST(P)$ as a boolean circuit. The input to the circuit will be the binary representations of the coordinates of the n points (each coordinate between 0 and G_n , requiring $L = \lceil \log_2(G_n + 1) \rceil$ bits). The circuit will compute distances (squared distances to avoid square roots, and compare squares to maintain integer arithmetic), compute the MST (using Kruskal’s algorithm implemented as a circuit), and compute the SMT by enumerating all possible Steiner points (which are also points of the same grid, because optimal Steiner points can be assumed to lie at intersections of grid lines? This requires a separate argument; we will assume we only need to consider Steiner points from the same grid, which is a standard technique). The circuit size will be constant for fixed n (though huge). The Tseitin transformation then gives a 3-SAT instance of constant size.

Remark 2.1. *The discretisation theorem is used only to justify that checking grid configurations is sufficient. In the formalisation, we will take the existence of the grid and the fact that any counterexample implies a grid counterexample as an axiom. The explicit construction is not needed for the logical proof; we only need the existence.*

3 Formalisation in Lean4

The Lean code imports the discretisation theorem as an axiom.

Listing 1: SeriesXXVII/Discretisation.lean (final)

```
1 import Mathlib.Data.Nat.Basic
2 import Mathlib.Data.Real.Basic
```

```

3 import Mathlib.Data.Fintype.Basic
4
5 open Real
6
7 -- For each n      N0, we have a grid size
8 noncomputable def grid_size (n :      ) :      := 2^n -- placeholder
9
10 -- Axiom: any counterexample can be moved to the grid
11 axiom discretisation (n :      ) (hn : n      N0) :
12   (P : Finset (      )) (h_card : P.card = n),
13   (SMT_ratio P < 2 / sqrt 3)
14   (Q : Finset (      )),
15   (      p      Q, 0      p.1      p.1      grid_size n      0      p.2
16     p.2      grid_size n)
17   Q.card = n
18   SMT_ratio (real_points_of_grid Q) < 2 / sqrt 3

```

4 Conclusion

We have established a discretisation theorem that reduces the continuous Gilbert–Pollak conjecture to a finite combinatorial search on a grid. The grid size depends on n and is constant for each fixed $n \leq N_0$. The next article (XXVII.2) will provide the asymptotic bound, and XXVII.3 will construct the boolean circuit for the grid configurations.

References

- [1] Du, D. Z., & Hwang, F. K. (1992). A proof of the Gilbert–Pollak conjecture on the Steiner ratio. *Algorithmica*, 7(1), 121–135.
- [2] Ivanov, A. O., & Tuzhilin, A. A. (2012). The Steiner ratio of the Euclidean plane. *ArXiv:1203.3122*.
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